Lab 2

COMPSCI 2XC3

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Summary

Experiment 1:

As the number of edges increases while the number of nodes remains fixed, the probability of a graph containing a cycle rises sharply from near 0 to near 1, showing a clear phase transition, and indicating that sufficiently dense graphs almost always contain cycles.

Experiment 2:

As the number of edges increases, the probability of a graph being connected rises rapidly from near 0 to near 1 after a critical threshold, showing that once enough edges are added to connect components, graphs quickly become fully connected.

Approximation Experiments:

Approx1 consistently performs the best and most reliably, with the lowest and most stable ratio. Approx2 and approx3 have similar but worse performance overall; approx2 tends to perform better in denser graphs, while approx3 is slightly more stable in some medium-density cases.

IS vs VC Analysis and Discussion:

For every tested graph, $|MVC| + |MIS| = |V|$

always holds, confirming that the complement of a minimum vertex cover forms a maximum independent set and validating the theoretical equivalence between MVC and MIS in undirected graphs.

Experiment 1

Input Data:

- nodes: 50
- runs = 200
- edges = range(0, 50, 1)

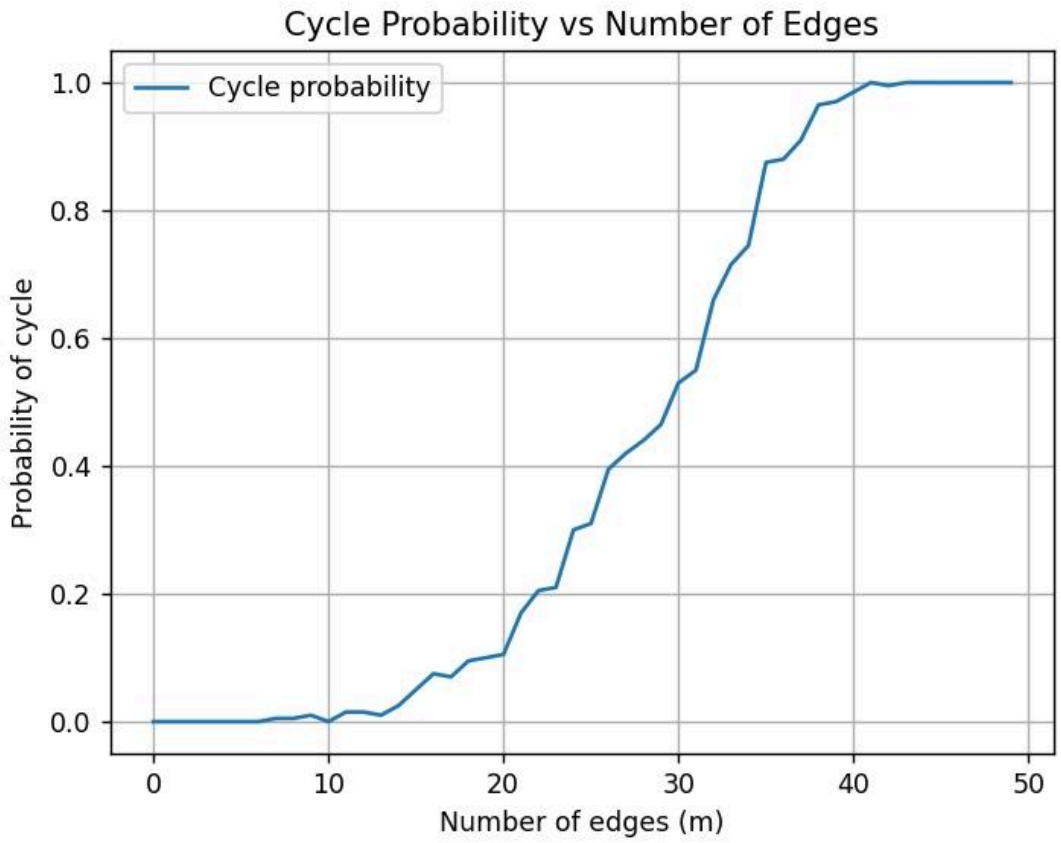

Figure: Cycle Probability vs Number of Edges

Conclusion:

This experiment investigates how the probability of a graph containing a cycle changes as the number of edges increases while keeping the number of nodes fixed. From the graph, we observe that when the number of edges is small, the probability of having a cycle is close to 0. As the number of edges increases, the probability rises sharply and quickly approaches 1. This demonstrates a clear phase transition behavior in random graphs. Once the graph becomes sufficiently dense, cycles almost always exist.

Experiment 2

Input Data:

- nodes: 50
- runs = 500
- edges = range(25, 250, 5)

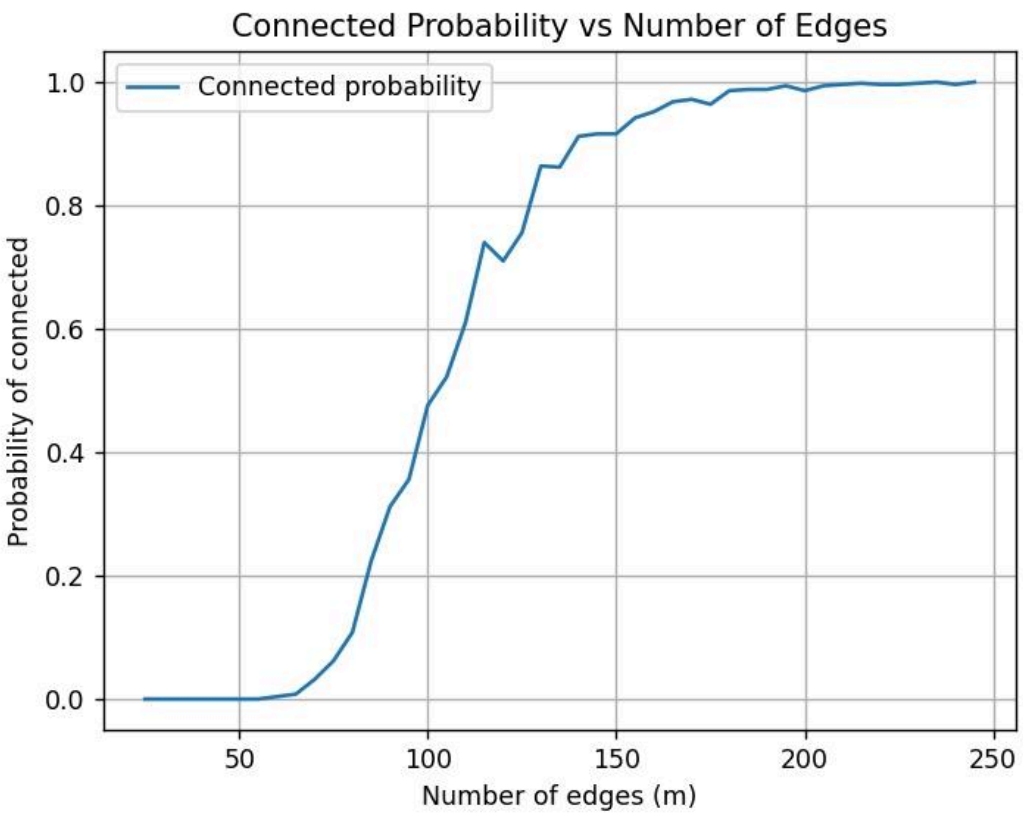

Figure: Connected Probability vs Number of Edges

Conclusion:

This experiment examines how the probability of a graph being connected changes as the number of edges increases. The results show that when the number of edges is small, the probability of connectivity is near 0. As more edges are added, the probability increases rapidly after a certain threshold. Eventually, the probability approaches 1, meaning almost all graphs become connected. This reflects the well-known theoretical result that connectivity emerges around a critical threshold of edges. Once enough edges are added to connect different components, the graph quickly becomes fully connected.

Approximation vs Number of Edges

Input Data:

- nodes: 10
- runs = 1000
- edges = 1, 5, 10, 15, 20, 25, 30, 35, 40, 45

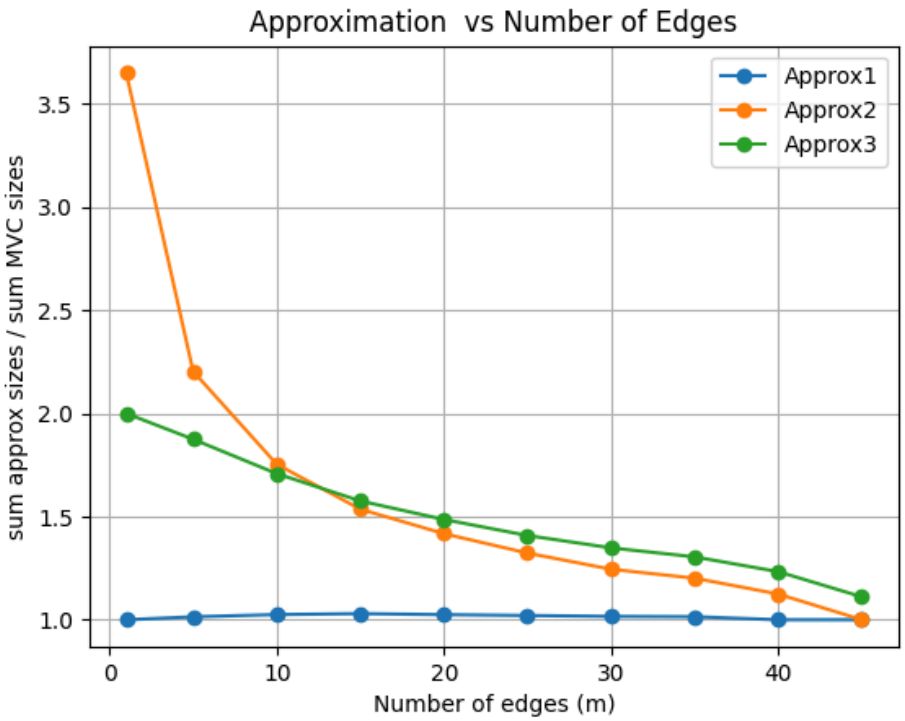

Figure: Approximation vs Number of Edges

Conclusion:

This experiment compares the approximations of three algorithms as the number of edges changes. As shown in the graph, approx2 exhibits an almost quadratic-like drop as the density of the graph increases. Its approximation outperforms approx3 when the number of edges reaches around 13, and it becomes nearly perfect when the graph is full. In contrast, the performance of approx3 shows a relatively linear decrease as the number of edges increases. Meanwhile, approx1 maintains nearly perfect performance across different numbers of edges, staying close to 1 with only very small fluctuations.

Approximation vs Number of Nodes

Input Data:

- nodes: 7,8,9,10,11,12,13,14,15
- runs = 1000
- edges = 20

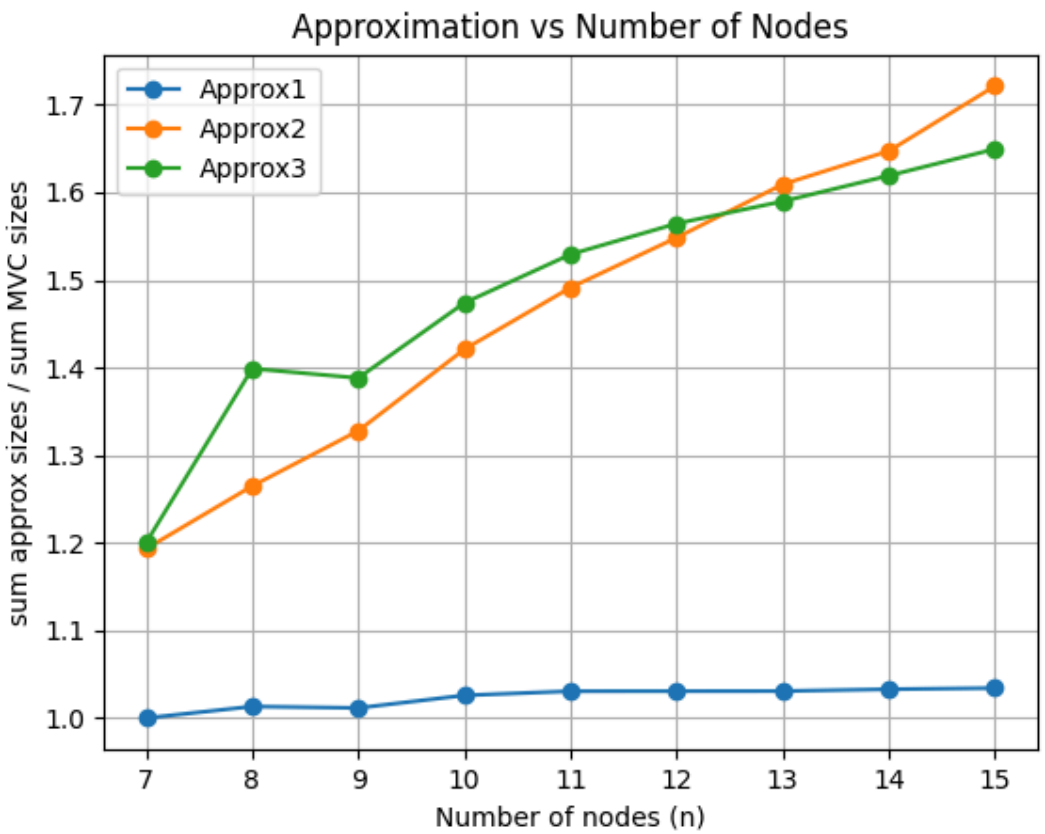

Figure: Approximation vs Number ofNodes

Conclusion:

This experiment compares the approximations of three algorithms as the number of nodes changes while keeping the number of edges constant. The performance of approx2 and approx3 exhibits a similar linear increase. Both have a relatively higher ratio than approx1, and approx2 is surpassed by approx3 at around 13 nodes. In contrast, approx1 shows much better performance and stability than the other two, with only a slight increase as the number of nodes increases.

Approximation vs Number of Edges

Input Data:

- nodes: 7,8,9,10,11,12,13,14,15
- runs = 1000
- density = 0.5

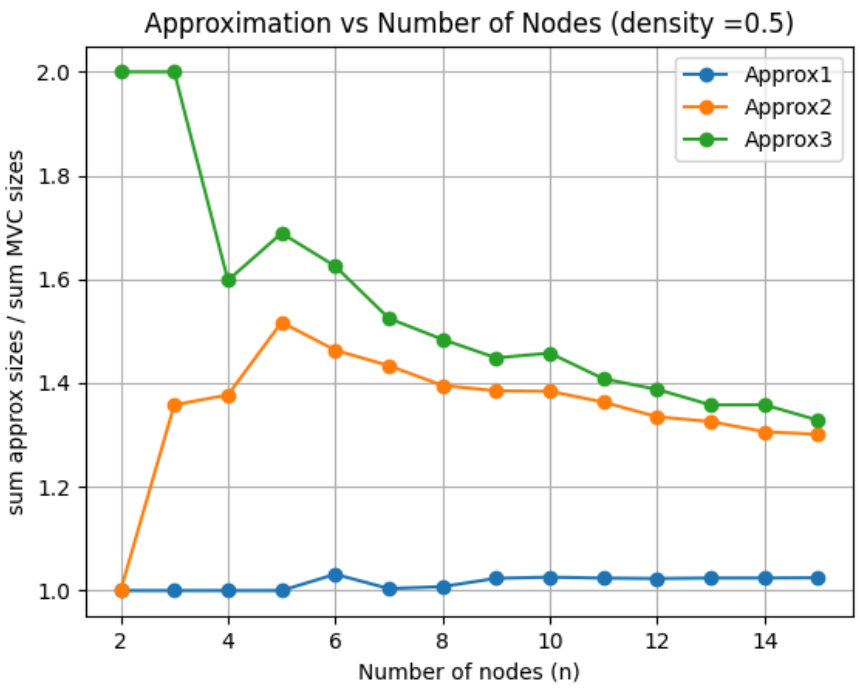

Figure: Approximation vs Number of Nodes(density = 0.5)

Conclusion:

This experiment compares the approximations of three algorithms as the number of nodes changes while keeping the density fixed at 0.5. As shown in the graph, approx2 first increases when the number of nodes is small, reaching a peak around 5 nodes, and then gradually decreases as n continues to grow. Approx3 shows a similar overall decreasing trend, although it starts from a higher ratio and decreases more steadily. In contrast, approx1 maintains an almost perfect performance across all values of n, staying very close to 1 with only small fluctuations.

Conclusion

From all graphs, approx1 consistently performs the best. Its ratio always stays between 1.00 and 1.1, with hardly any fluctuations. Its performance is consistently better than the other two, so we can say it is the best among all three.

For approx2 and approx3, their performance is clearly worse than approx1, but they are relatively close to each other. When the density is low , approx2 sometimes performs worse at very small graph sizes, but as the number of edges increases, approx2 improves quickly and can outperform approx3 in denser graphs. In contrast, approx3 performs relatively better in some medium-density cases and shows a more stable decreasing trend as the graph becomes larger. Under fixed density with increasing n, both approx2 and approx3 gradually improve and become more stable.

Overall, approx1 is clearly the best and most reliable algorithm. Approx2 and approx3 are comparable in general, but approx2 tends to perform better in denser graphs, while approx3 can be slightly more stable in some medium-density situations.

IS vs VC Analysis and Discussion

In this experiment, we generated multiple random graphs and computed both the Minimum Vertex Cover (MVC) and the Maximum Independent Set (MIS) for each graph. For every graph tested, we observed that:

$$|MVC| + |MIS| = |V|$$

where $|V|$ is the number of vertices in the graph.

This empirical result confirms a fundamental theorem in graph theory: the complement of a minimum vertex cover forms a maximum independent set. Intuitively, since a vertex cover contains at least one endpoint of every edge, the remaining vertices cannot share an edge and therefore form an independent set.

Across all tested graph sizes and edge densities, this relationship consistently held. Our experimental results therefore validate the theoretical equivalence between Minimum Vertex Cover and Maximum Independent Set in undirected graphs.

Appedix

Experiment 1: **experiment1.py**

Experiment 2: **experiment2.py**

Approximation Experiments: **approximations_experiments.py**